

Predicting any observed quantity of the Earth: a design for a global ocean–land–atmosphere model that reads the observational record through a learnable space–time cone

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Abstract

We specify a two-stage model that predicts any observed quantity of the Earth system — a grid cell, a station, a float profile, a satellite pixel — at any place, time and footprint, from the whole observational record of ocean, land and atmosphere. The design rests on one fact: a quantity at a place and time depends on a cone in space–time whose reach grows with the speed of the fastest signal carrying information about it and ends with that signal’s memory, and it influences the mirror of that cone into the future. Both stages sample this hourglass. Stage 1 reads raw values over the last thirty days through a waist of several cells and writes one embedding per cell and pentad. Stage 2 reads those embeddings over two years at log-spaced lags and writes the embeddings of the next year as a residual on a contractive linear propagator. Every value enters through one token — value, observed flag, offset from the anchor, footprint, source — so dense 0.25° grids, kilometre-scale fields, tile catalogues and point observations share one encoder without resampling; every prediction leaves through one query-based decoder that takes the same coordinates and footprint. The cone’s shape — a drift toward the source, a waist ellipse, a growth ellipse, a scale and a memory — is learned per 1° cell and per channel under guards that stop the model moving its own questions to easy water. We state the data, the geometry, both stages, the training ladder, the evaluation protocol with its null ladder and the comparisons that decide each choice, with verdict rules fixed before any run. A results section is reserved with the tables it will hold.

1 Introduction

We specify a model that, given the observational record up to time t , returns the value an instrument would read at place $\mathbf{x}$, time $t+\ell$, depth or height, and footprint (the area and span the reading averages), for any channel the record has contained. A transport through a mooring section, an El Niño index, a coastal pixel’s chlorophyll, a station’s temperature, a gauge’s sea level, a forest’s canopy height at a laser footprint and a tower’s carbon flux are instances of one query, answered by one representation and one decoder. The Atlantic overturning, the ocean carbon sink, El Niño, sea-surface temperature and the land biosphere are the five read-outs tracked, because each has a long truth series; they are properties of the instruments, not of the target.

The design rests on one fact, stated in §2.1: what can bear on a value at a place and time lies in a cone in space–time, and the value bears on the mirror of that cone (Figure 1). The record at 0.25° and five-day cadence is of order 10^9 values a year, the kilometre tier a thousand times more, and point observations are irregular in every coordinate; no transformer attends over that directly. Stage 1, the codec, reads raw values in a small cone — thirty days, hundreds to a few thousand kilometres — and writes one embedding per cell and pentad; it is the only stage that sees a raw value. Stage 2, the forecaster, reads embeddings in a large cone — 35 days to two years, up to the planetary scale — and writes the embeddings of the future cone; it is what is rolled forward and scored. Stage 1’s hourglass is the small solid at the centre of stage 2’s. The split is in time, not distance.

Premises. (i) At one basin’s data volume the forecast is data-limited — the held-out optimum arrives early and width does not move it — so the design grows data before parameters, and each rung of the ladder is bought on the one below. (ii) A linear inverse model in embedding space and damped persistence are strong beyond two weeks, so the learned forecaster is a residual on a contractive linear propagator and is reported against both. (iii) A read-out with few effective degrees of freedom cannot

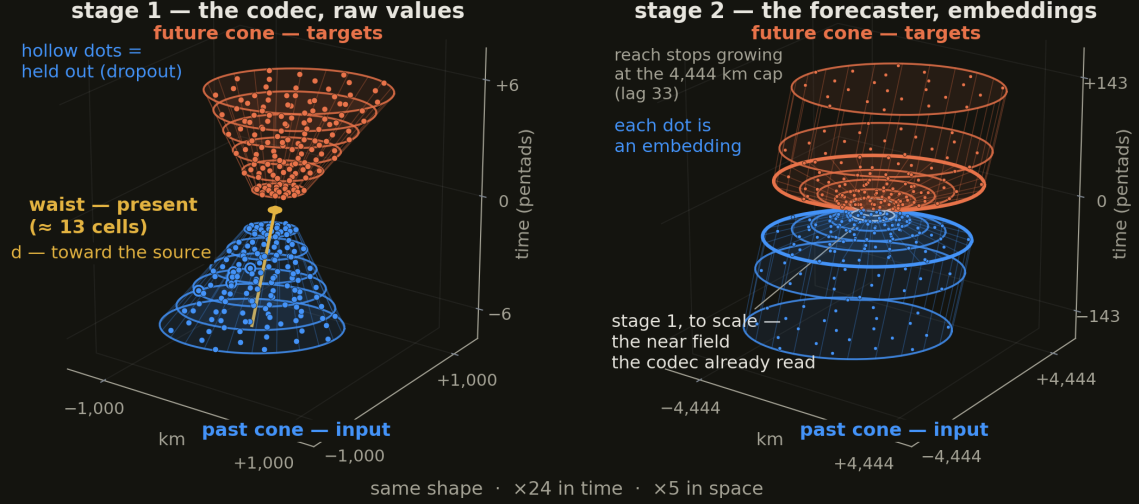


Figure 1: The hourglass at both scales. Each lag is a 24-point sunflower (phase-rotated by the golden angle per lag) inside an ellipse that grows with lag; past half input, waist present, future half target only. Left: stage 1 over raw values, six pentads each way, ≈ 900 km at six pentads for the ocean surface, a thirteen-cell waist. Right: stage 2 over embeddings, 7–143 pentads each way, to a 4,444 km cap; stage 1’s hourglass is the small solid at the centre.

resolve a design choice; comparisons are decided on the dense field. (iv) The holdout is terminal and spatial, and the exclusion of held-out data from every window the forward pass can see is certified before training. A technical report on the North Atlantic pilot [1] is the dated record behind these premises; this paper does not repeat it.

Section 2 states the method: the problem, the token every observation becomes, the cone and its learnable geometry, the two stages and the query decoder. Section 3 describes the data, Section 4 the training ladder, and Section 5 the evaluation protocol and the comparisons that decide each design choice, with verdict rules fixed before any run. Section 6 reserves the results; Section 7 places the design among existing models.

2 Method

2.1 Problem formulation

Queries. Let $\mathcal{O}_{\leq t}$ be every observation made up to time t : gridded fields, point and profile measurements, image tiles, each a value with a place, a time, a depth or height and a footprint. A query $q = (\mathbf{x}, t + \ell, z, c, f_p, f_t)$ names a place, a lead $\ell \geq 0$, a depth or height, a channel c and a footprint in space and time. The model returns a predictive distribution $p(y_q | \mathcal{O}_{\leq t})$ — a mean and a variance in the channel’s transformed space (§2.5) — for any query whose channel and footprint occur among the training targets. Lead zero with the value hidden is a nowcast. Every stage below serves this one function.

The dependency cone. A driver moving at speed v reaches $\mathbf{x}$ at time t from within $v \Delta t$ at $t - \Delta t$. The points that can bear on $(\mathbf{x}, t)$ form a cone opening into the past, its reach set by the fastest mechanism carrying that channel and cut off by the channel’s memory: wind stress forgets in days, sea-surface temperature anomalies in months, a deep water mass in years. Each channel has a family (v, τ) . Run forward, the same argument gives the points $(\mathbf{x}, t)$ can bear on: the point mirror of the past cone. Past cone, waist and future cone are an hourglass (Figure 1). Advection is what the mirror looks like: water came from upstream and goes downstream.

2.2 One token per observation

The encoder attends over places at times, each carrying every channel there:

$$\text{token} = (\mathbf{v}[C], \mathbf{m}[C], \Delta x, \Delta y, \Delta z, \Delta t, \log_2 f_p, \log_2 f_t, n_R, s, \text{sphere}, \text{season}). \quad (1)$$

$\mathbf{v}$: standardised anomalies of the group’s channels; $\mathbf{m}$: observed flags; offsets from the anchor in km, m and days; footprint $\log_2 f_p = \log_2(\text{km}/27.83)$ and $\log_2 f_t = \log_2(\text{days}/5)$, clamped to ± 12 (a Sentinel-2 pixel $(-11.4, -4)$, a laser footprint $(-10.1, -12)$, a 30 m annual map pixel $(-9.9, +6.2)$, an Argo profile $(-4, -4)$, a 4 km daily cell $(-2.8, -2.3)$, an OISST pentad cell $(0, 0)$, a reanalysis cell at its true support $(+2.9, 0)$, a monthly 1° Argo field $(+2, +2.6)$); n_R : how many such observations were within reach; s : a learned source embedding; sphere code; day-of-year harmonics. One token per location and lag, not per channel: adding a channel adds a vector entry, not a token. Footprint is constant per group for dense sources and per row for sparse ones; nothing else depends on the storage tier (§3.2), so the encoder never learns which file a value came from. A 1° value is one token at the cell centre, seen from sixteen offsets by the sixteen anchors in it. Point observations enter where and when they were made: the k nearest in space–time within the cone, k small (three to eight per depth group), one-sided in time, padded with *miss* tokens. A value the encoder may not see carries a *hidden* marker, distinct from *never observed*.

2.3 Sampling the space–time cone

Waist. The present is a disc of radius two cells (≈ 13 cells, a small sunflower), read and reconstructed; the embedding z is written for its centre. Gradients need neighbours: thermal wind is a density gradient, geostrophy a height slope, and neither exists in one cell. The waist carries the cone’s shape at lag zero, so a jet can have an elongated waist.

Past cone. Lags 1–6 pentads, every pentad: five days is the tensor’s cadence and thirty days is the codec’s job. At each lag the sampling set is a 24-point Vogel sunflower $\mathbf{u}_i$ mapped through a Gaussian footprint,

$$\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v, \quad p(\ell, i) = \mathbf{x}_{\text{anchor}} + \ell \mathbf{d} + \Sigma(\ell)^{1/2} \mathbf{u}_i, \quad (2)$$

with drift $\mathbf{d}$ per pentad toward the source, waist ellipse Σ_0 and growth ellipse Σ_v (standard deviation linear in lag). The sunflower’s phase advances by the golden angle per lag, so the six slices interleave. The anchor column is sampled at every lag.

Far ring. Every lag also keeps a fixed ring of 24 points at log-spaced radii from 28 km to the antipode, read at full weight and never gated. The design speed is the fastest measured mechanism of the family times 1.5 (the ocean: 3.6 ms^{-1} , so 5.4 ms^{-1} and 2,333 km per pentad; wind stress: the atmosphere, global within a pentad). The ellipse gives direction and density; the ring guarantees no driver is excluded by construction.

Prediction cone. The past cone point-mirrored: centre $-\ell \mathbf{d}$, same $\Sigma(\ell)$, leads +1 to +6, sampled the same way, read by nobody. Each point is a decoder query and a target. The anchor column at every lead is always a target.

2.4 Learnable per-channel geometry

Per-channel geometry. What moves a signal differs by channel even inside one file group: currents follow the flow; sea-surface height anomalies move west with eddies and Rossby waves whatever the current does [3]; sea ice moves with the wind at 1–2% of its speed and to its right [4]; sea-surface temperature follows the wind at short lags and the current at long ones [5]; below ~ 800 m at the western boundary the deep current runs opposite to the upper limb [6]. So every geometry number is per channel: $\mathbf{d}_c$, $\Sigma_{0,c}$, $\Sigma_{v,c}$, a log-scale s_c and a memory τ_c (two scales for the L-shaped channels, lags 0–1 and 2–6). They are stored hierarchically — a group value plus a per-channel residual under L2 shrinkage — so a channel that shares its group’s carrier costs nothing and one that does not pays for its departure with evidence. Initial values come from a table set by physics: wind stress 500 km with a ten-day memory; currents, height and profiles 130 km per pentad with a long memory; SST and mixed layer $\max(\text{that}, 500 \text{ km})$ at lags ≤ 1 then the ocean’s growth; land 400 km flat, no drift; $\mathbf{d} = 0$

everywhere; the waist ellipse at the measured flow anisotropy 0.71. Step zero is the fixed hourglass. The biosphere channels are land channels: no carrier of their own, an aperture that opens over the atmospheric and soil tokens, fire the one exception with the wind as its carrier at 5–30 km a day. Their memories — weeks for leaf area, decades for canopy height — exceed the cone, so a slow channel is read as a state: the previous annual map is an input token at lag ≥ 73 pentads and next year’s map at the same footprint is the target; the same hourglass at the annual time footprint.

Aperture, at zero extra tokens. A token carries all channels of its group, so each channel reads the shared points through its own weight

$$w_{c,i,\ell} = \exp\left(-\frac{1}{2}(p_i - \ell \mathbf{d}_c)^\top \Sigma_c(\ell)^{-1}(p_i - \ell \mathbf{d}_c)\right) e^{-\ell \Delta t / \tau_c}, \quad (3)$$

applied to that channel’s value and flag entries before projection. A point outside channel c ’s aperture in space or memory contributes ≈ 0 for c and full weight for the others (400 km at lag 1: currents 0.30, SST 0.73 at the initial apertures). The weight is smooth in every geometry number, so the loss reaches all of them through every point at once. The same weight applies to tier-P tokens, whose offsets and depths are known: each pressure-level channel gets its own direction over the same k nearest profiles, with no sampling ellipse for profiles. The ring is not gated.

Parameterisation. $\mathbf{d} = \tanh(\mathbf{x}) v_{\text{design}} \Delta t$, capped by construction. Ellipses as matrix logarithms, $S = \log \Sigma = m\mathbf{I} + k \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$: no kilometres, no angle to wrap, a circle is zero, and an Adam step moves every radius by the same fraction of itself [7]. Floors (waist one cell, Σ_v the design speed) are softplus offsets in log space, not clips. The numbers live on a 1° map per channel (64,800 cells; bilinear blend at the anchor; $\approx 34\text{M}$ nominal, half live, no compute), because the currents that carry the overturning are $\sim 100\text{km}$ wide and a 10° cell would learn a drift near zero. Each map is a coarse 10° map plus a 1° residual under L2 and Laplacian priors, so a data-poor cell inherits its region. A first-harmonic seasonal drift is a second-phase option for monsoon regimes.

Learning: gate first, harden later. In phase one the candidate points are fixed — a log-radial set of 36 per lag to the design reach plus the anchor column — so reads are regular and independent of live weights, and the geometry learns through the aperture alone. Every 2,000 steps the dense table per region is re-baked, the 24 points per lag drawn from the group’s mixture of channel ellipses, and swapped at an epoch boundary. Geometry parameters get ten times the codec’s learning rate [8]. Reading at continuous positions was rejected: it ties the loader to live weights and its gradient sees four cells.

Guards. A movable question moves to easy water. So: the loss reaches the geometry only through input reads (target positions carry a stop-gradient); the future cone is tied to the past cone by the mirror, per channel; evaluation geometry is frozen to the fixed hourglass for every arm; the aperture never enters the loss weights; the ring stays ungated. The residual effect — mirrored targets move with the input cone — is monitored: every evaluation reports target variance under the learned against the fixed cone, and a drop over 10% is a flag.

2.5 Stage 1: encoding observations

Tasks. Each example is one anchor at one bin and draws one task (Table 1). Dropout patterns fire independently: one channel hidden at the waist; lags $\leq \ell_0 \in \{1, 2, 3\}$ hidden; a 90° bearing wedge hidden; points hidden at random ($p = 0.3$); for the nowcast, the whole waist. The task is not signalled by a token; hidden markers and the lead in each query suffice. Point observations — a profile, a laser footprint, a fluorescence sounding, a tower’s half-hour — are hidden and predicted like cells; an annual map is a T1 or T4 target at its own footprint and year, with the previous year’s map visible. Three rules are not knobs: a target’s channel must be visible somewhere in the input (the sampler counts violations, required zero); the waist family scores hidden channels only; and every (task, family, lead) is reported against the score of predicting the mean, and is called learned only where the ratio is below one held out.

task	input	targets	share	what it teaches
T1 forecast	waist + past cone	future cone	0.35	the full problem
T2 snapshot forecast	waist only	future cone	0.15	what history buys, inside one model
T3 nowcast	past cone under dropout; waist hidden	waist	0.25	the present from the past
T4 fill-in	waist + past cone under dropout	hidden cells and points	0.25	reconstruction from partial observation

Table 1: Stage-1 tasks, drawn per example. Shares are a starting point and the first thing §5.2 varies.

Model. A strict masked autoencoder [9]: the encoder sees visible tokens and entries only, never a placeholder. It is a transformer over the example’s tokens — about 900 in the gate phase and 1,200 after hardening (six lags $\times$ column, ellipse and ring $\times$ four groups, plus waist and profiles) — with Perceiver-style cross-attention into a fixed latent set [10], writing z of width 128–256. The bottleneck is the token count, not the width. The decoder reads z alone: a query (offset, channel, sphere, depth, lead, footprint) cross-attends to z and emits a mean and log-variance. A decoder that could read the encoder’s latents would let z carry nothing; that path is closed. At the larger rungs the feed-forward block is a mixture of experts routed by sphere and channel group [14].

Objective. A heteroscedastic Gaussian per channel group in its transformed space, decoupled:

$$\mathcal{L} = \|\mu - y\|^2 + \frac{1}{2}e^{-s}(y - \text{sg}[\mu])^2 + \frac{1}{2}s, \quad s = \log \sigma^2, \quad (4)$$

so the mean trains by squared error and the variance by the likelihood with the mean detached, which removes the variance head’s habit of explaining a hard target away [11]. $s \in [-8, 8]$; the first 2,000 steps run at $\sigma = 1$; Gaussian in log space for log-normal channels, a hurdle (wet/dry $\times$ log-Gaussian) for rain, logit-Gaussian for fractions. Coverage of the $\pm 1\sigma$, $\pm 2\sigma$ bands is the calibration check.

Weighting across targets. A plain sum over targets whose residual errors differ by orders of magnitude lets the hardest dominate. Standardising weights by marginal variance, which conflates *varies a lot* with *hard to predict*. The right scale is the irreducible error:

$$\mathcal{L}_\mu = \sum_k w_k \text{MSE}_k, \quad w_k = 1/\text{MSE}_k^{\text{ref}}, \quad \text{no gradient through } w_k, \quad (5)$$

over classes $k = (\text{channel, family, lead})$. This is the stop-gradient form of $\sum_k \log \text{MSE}_k$ — Gaussian likelihood with one fixed variance per class — so each class contributes its relative improvement, and the residual scale stays out of the co-adapting σ head. The reference is the held-out class error, never the final training error. At a new tier it is an exponential moving average under stop-gradient, started from the marginal variance and annealed over the first thousands of steps; at a tier with a previous run the weights are frozen from it and recomputed after, a run being repeated once if any moved by more than $2\times$. Heavy-tailed classes (rain, chlorophyll) use a robust scale or a Huber error. Logged per class: $w_k \text{MSE}_k$ (near one at convergence) and $1 - \text{MSE}_k/\text{var}_k$. An explicit multiplier for a class a use cares about sits on top, kept separate.

Stability. Collapse is invisible to reconstruction loss, so the effective rank of z on a fixed held-out batch and a scale-invariant probe correlation are logged each evaluation; two sub-threshold readings abort. No quantisation from random initialisation. Gradient clip 1.0; AdamW, $\beta_2 = 0.95$, weight decay 0.1, 2,000-step warm-up, cosine to 10%; bf16 with fp32 master weights; stop at the held-out minimum, at most four epochs [12].

2.6 Stage 2: forecasting embeddings

Inputs. Tokens are the codec’s embeddings: mean and log-variance of z , offset, statics and season at each sampled cell and lag. The past cone reads lag 0 and lags 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 pentads — twelve slices, not 144 — each with the anchor column, a 24-point footprint whose reach grows at 130 km per pentad ($\approx 1,000$ km at lag 7, the 4,444 km cap from lag 33) and a ring to the antipode: about 600 tokens against twenty thousand for a dense window, which is the ablation. The same per-channel geometry, maps and guards apply.

Linear backbone. For the anchor column,

$$\hat{z}_{t+\ell} = \mathbf{G}_\ell z_t + f_\theta(\text{cone}, \ell), \quad \mathbf{G}_\ell = \mathbf{U} \text{diag}(e^{-\ell/\tau_j}) \mathbf{U}^{-1}, \quad \tau_j > 0, \quad (6)$$

a contractive propagator with learned time constants, initialised from a linear inverse model [15] on training-year embeddings, plus a transformer f_θ with a zero-initialised output projection. At step zero the forecaster is the embedding-space LIM; it must earn its distance from it held out. Damped persistence is the diagonal case and its own null.

Direct multi-lead targets. The mirrored future cone at leads 1, 2, 3, 5, 8, 13, 21, 34, 55, 73 pentads, each a query to the same head, predicted directly, not by chaining: no compounding, amplitude set by the data at that lead, targets where the correlation still changes. Anchor column weight 1, footprint 0.25; the footprint targets make the prediction a coherent field. Beyond a year the predicted anchor embeddings feed back as the recent past: the multi-year roll is a mode, not the objective.

Objective. The decoupled Gaussian on each target embedding, weighted per class by the rule of §2.5 starting from the predict-the-mean bar; plus a decoded-space squared error through the frozen codec decoder at leads 1, 8, 34 for the scoreable channels, because the embedding metric is not the scored metric; plus, for the multi-year mode, a two-step pushforward with a stop-gradient on the first step. Under a squared-error mean the amplitude shrinks with correlation, so calibration is a property of the held-out minimum; the identity $\text{MSSS}_{\text{clim}} = 1 - (1 + a^2 - 2a \text{ACC})$ is logged per lead as the check, and a scaling fitted on training-year starts is a diagnostic only.

Regularisation. The globe supplies every regime; the log-spaced cone cuts parameters per example seventyfold; the free part is a residual on a backbone the data pin down. Then: Gaussian noise on input embeddings at 0.7 of the persistence scale; dropout of past points; weight decay; early stopping; and a seed ensemble of three to five whose spread is the first probabilistic forecast. Once the deterministic level exists, a shared low-dimensional noise vector conditions the head so samples cohere across cells and leads, scored by fair CRPS [18] — trained on marginals, such an ensemble already reproduces joint spatial structure [21]; categorical marginals over a quantised embedding are the alternative where the conditional is two-humped, as an El Niño index at 6–24 months may be.

2.7 Decoding queries

A prediction is a query in the token’s own coordinates: offset in space, depth, time; channel; sphere; source; footprint. A 0.25° cell over a pentad, a 300 m pixel on a day, a station at an hour, a float at its depths, a canopy’s height at a 25 m footprint and a 30 m pixel’s loss year are one decoder, because it was trained on targets with those coordinates and footprints, each scored against the observation that has that support. The domain is what the training targets contained, at the footprints they came with; a channel never observed cannot be asked for, and a footprint far outside the seen range is an extrapolation and is reported as one. A query where the source never covered — SST over land — returns a variance that says so, because the variance head was never rewarded for confidence there. A functional of many cells — a transport, an area index, a warm-water volume — is an attention head over the predicted embeddings of its cells, trained on training-year true embeddings, never on a spatial mean (geostrophic transport is the contrast across a section, and a mean cancels it). Read-outs are the cheap end of the system and choose between no models. A point store’s own record verifies its forecast; where a licence forbids redistribution, the score is published and the values are not.

3 Data

Nothing is resampled to a common grid. Each source is stored at its own resolution and cadence in one of three tiers; the sampler turns all of them into one token at read time; the decoder is queried at a location and a footprint.

3.1 The 0.25° pentad grid

Grid and time. A point-aligned 0.25° grid, $721 \times 1,440$ cells, rows south first, longitude wrapping; a five-day bin (pentad) from 1982-01-01, bin b covering days $[5b, 5b + 5)$; 3,142 bins to 2024-12. A bin is an averaging window, not a date. A coarse 1° grid (181×360) is defined the same way; a coarse channel is read at its nearest coarse point, so sixteen fine cells see one number. That is the model’s honest view and is not interpolated.

Groups. Each group is an array $[T, H, W, C]$ in float16, bin-major, at its native grid and cadence:

- **ocean surface**, 0.25°, every pentad: current speed and its two components, sea-surface height and log mixed-layer depth (GLORYS, 1993–), sea-surface temperature and sea-ice concentration (OISST, 1982–);
- **atmosphere and land**, 1°, every pentad, every surface: wind stress and its within-pentad standard deviation (four), 2 m air temperature, 10 m wind (two), surface pressure, log precipitation rate, log snow water equivalent, soil moisture and temperature (0–10 cm), latent and sensible heat flux, skin temperature (NCEP; ERA5 is the intended replacement);
- **ocean interior**, 1°, monthly: temperature and salinity at sixteen pressure levels, 10–1,900 dbar (Roemmich–Gilson Argo, 2004–), each month in the pentad holding its fifteenth;
- **ocean colour**, 0.25°, from 1997-09: block mean of $\log_{10}$ chlorophyll over the 6×6 OC-CCI 4 km cells under a point, and the observed fraction of cell-days — the evidence behind the value;
- **statics**: surface class (702,642 ocean, 226,495 land, 107,074 ice sheet, 2,029 inland water; ice over ocean on a shelf) and elevation.

No channel is masked by sphere: a land cell reads its skin and air temperature and soil, an ocean cell its surface fields, every cell the shared atmosphere. A channel is a quantity and an instrument, never a sphere; observed sea-surface and land-surface temperatures are separate channels.

Standardisation, transforms, missing values. Stored values are $x' = (x - \mu)/\sigma$ per channel. At load time every channel becomes a standardised anomaly against a day-of-year harmonic climatology (annual and semi-annual) fitted on training years only, over a training-year standard deviation; no held-out year touches any statistic. Log-normal quantities (mixed-layer depth, chlorophyll, precipitation, snow) are logged first; bounded fractions (sea ice, soil moisture) go through a logit. A value that does not exist — a land cell’s SST, a month without Argo, a cell under cloud — is not zero: every token carries an observed flag per channel, and an unobserved entry is masked from loss and input. Holes in the grid are holes in the sources, counted per channel and bin at build time.

3.2 Finer sources in three storage tiers

Three facts decide how finer sources enter. A coarse value copied into fine cells is a lie: nine copies are not nine measurements. A fine value averaged into a coarse cell discards what the source was bought for: 4 km colour into 0.25° keeps one part in thirty-six. And missing is not damage, nor is far away: the nearest profile’s value depends on its distance, which a grid cell hides. Sources are therefore stored by density, in three tiers under one registry:

- **G, gridded dense** — fields at $\lesssim 5$ km, daily to monthly, near-complete: colour at 4 km, SST at 5.6 km per sensor and 2.2 km merged, sea ice at 6.25 km, 4 km cloud-top temperature, 0.1° daily rain, the daily 0.05° land temperature, snow and reflectance, 500 m leaf area every eight days, and the annual maps — forest loss, canopy height and biomass, the leaf calendar — at 30–500 m. A group finer than 0.25° is sharded: one compressed file per group and pentad, one frame per day (or three hours, or one per year), 256×256 tiles with a byte index, so a read fetches the tiles around the anchor and an empty tile costs nothing. Footprint is a constant per group.
- **P, points, profiles, tracks** — Argo and BGC-Argo, every non-Argo profile since 1772, ship and buoy reports since 1662, bottle chemistry, drifters, tropical and open-ocean moorings, the overturning arrays’ instruments, surface CO₂, along-track altimetry at 7 km and SWOT swath

store	what it is	producer	space	time	record	tier
seaice-asi	daily sea-ice concentration, microwave	Univ. Bremen, AMSR-E/2	6.25 km polar	daily	2002-11, 2012-	G
oc4k	daily chlorophyll, attenuation, reflectance	ESA OC-CCI v6.0	4.6 km	daily	1997-09-	G
sst-cci05	daily SST per sensor	ESA SST CCI v3.0	5.6 km	daily, day/night	1979-07-	G
sst-acspo02	daily SST merged across sensors	NOAA ACSP0	2.2 km	daily, AM/PM	2000-02-	G
irtb	geostationary cloud-top temperature	NOAA CPC	4 km	30 min (3-hourly first)	1998-01-	G
precip01	daily satellite rain (IMERG)	NASA/JAXA GPM	11 km	daily	2000-	G
wind12	scatterometer ocean wind	OSI SAF; QuikSCAT	12.5 km swath cells	per pass	1999-2009, 2007-	P
swot	sea-surface height, two 50 km swaths	NASA/CNES SWOT	2 km	per pass, 20.86 d	2023-07-	P
swh	wave height along altimeter tracks	ESA CCI Sea State	≈7 km	1 Hz	1991-08-2023-12	P
icoads	ship and buoy weather reports	NCEI / NCAR, ICOADS 3.0	point	per report	1662-	P
wod	T, S, O ₂ profiles, Argo excluded	NCEI, WOD	profile	per cast	1772-	P
glodap	bottle carbon, nutrients, oxygen	GLODAPv3	bottle	per cruise	1972-2023	P
bgcarg	floats: O ₂ , nitrate, pH, chlorophyll	Argo GDAC	profile	≈10 d per float	2002-	P
oceansites	open-ocean moorings	OceanSITES	mooring	hourly or finer	1980s-	P
deep_arrays	the four overturning arrays' instruments	RAPID, OSNAP, SAMBA, MOVE	mooring	15 min-1 h	2000/04/09/14-	P
xco2	column CO ₂ soundings	OCO-2/3; GOSAT	< 3 km ²	per sounding	2014-06-	P

Table 2: The global fine-granularity family. Each store is kept at its own resolution and cadence and enters through the token of §2.2 with its own footprint. A one-month test fetch through the real builder replaces every size estimate before a store is built.

cells at 2 km, column CO₂ soundings, wave height, scatterometer wind cells; on land, stations, balloon soundings, tide gauges, buoys, gliders, flux towers, fire detections, rivers, soil probes, laser footprints of the canopy, fluorescence soundings, species occurrences. A store is a set of columns — time in integer seconds, latitude, longitude, values, mask, quality, hashed platform, per-row footprint — sorted by pentad with an index per bin, so the k nearest observations in space-time are one bounded scan. Pre-1982 rows keep negative bins.

- **T, tiles** — imagery at ≤ 1 km (Sentinel-1/2, Landsat, NISAR, OLCI, SLSTR, VIIRS, MODIS, the L- and P-band radar mosaics) and the 10 m and 1 m canopy-height maps are not stored as values. The store is a scene-date catalogue (footprint, sensor, time, cloud fraction, location) and a token cache: a local codec turns one tile-date into a few 64-dimensional vectors, stored as a tier-P store whose values are those vectors, footprint the tile and the day. AlphaEarth’s published embedding [2] — 64 numbers per 10 m of land per year — enters by the same path as a ready-made cache, so the programme’s own tile codec can be measured against it with everything else fixed.

One time axis (the pentad bin; tier-P rows keep their exact time, so Δt is fractional); one registry per family (tier, layout, cadence, channels, footprint constants, first bin, checksums, sources, licence, builder commit — a consumer dispatches on the tier); one sampler contract (`gather(anchor, family)` returns tokens per group under that group’s cone family, with a fixed slot count so batches stay rectangular). Stores whose licences forbid redistribution live on a private track in the same format, listed in the public registry.

The global fine-granularity family (Table 2) is the instance of tiers G and P over the ocean and air at ≤ 10 km and ≤ 5 days; a land-and-coast family holds the station networks, gauges, towers, fires, rivers, the 0.05° land fields, the imagery catalogue, the land-change targets and the biosphere stores of Table 3; a third is that family with the catalogue replaced by the AlphaEarth cache. Admission: raw measurements first, a derived product only where it carries measurements nothing else does or is itself a target a user will ask for (the annual maps), no gap-filled analyses; four named exceptions slightly coarser than the rule (12.5 km scatterometer wind, 11 km rain, one 10.5 km CO₂ sounder, eight-day leaf area); and every store must lower the error on one of the five read-outs (§5.2) or be dropped.

The local codec. A 64×64 tile at 10 m is 4,096 pixels of 13 bands and a 0.25° cell holds $\sim 7,000$ of them; they cannot be sampling points. The local codec is a masked autoencoder over one tile-date, conditioned on sensor metadata (resolution, incidence or sun angle) with a learned source embedding

store	what it is	producer	space	time	record	tier
gedi	laser relative-height profile, cover, biomass per footprint	NASA GEDI, ISS	25 m every 60 m	per shot	2019-04–	P
icesat2	laser terrain and canopy height per segment	NASA ICESat-2 ATL08	100 m	per pass	2018-10–	P
sif	chlorophyll fluorescence soundings	TROPOMI; OCO-2/3	7×3.5 km; 1.3×2.2 km	per sounding	2018-05–; 2014-09–	P
flux	tower CO ₂ , water and heat exchange	FLUXNET, 700 sites	tower	half-hourly	1991–	P
fire	active-fire detections	FIRMS, MODIS/VIIRS	375 m–1 km	per overpass	2000-11–	P
gbif	species occurrences, taxon as platform	GBIF snapshot	per-record uncertainty	day or month	1600s–	P
lai500	leaf area, absorbed light	MODIS/VIIRS	500 m	8-day	2000-02–	G
pheno500	green-up to dormancy dates, amplitude	MODIS	500 m	annual	2001–2025	G
canopy30	canopy height 2020	MCD12Q2	30 m	one year	2020	G
biomass100	above-ground biomass and its error	GLAD, Landsat fitted to GEDI	100 m; 1 km	annual	2005–2024	G
lossyear	the year each pixel lost its forest; tree cover 2000	ESA CCI v7; GEDI L4B	30 m	annual	1990/2001–	G
canopy_ref	10 m and 1 m canopy height, catalogued	Hansen GFC; JRC TMF	10 m; 1 m	one year; a decade's composite	2020; 2009–20	T
cat_palsar	L- and P-band radar, catalogued	JAXA PALSAR-1/2; ESA BIOMASS	25 m; 50–200 m	annual; per pass	2007–; 2025–	T
pace4k	hyperspectral colour and phytoplankton groups	NASA PACE OCI	4 km	daily	2024-03–	G

Table 3: The biosphere stores of the land-and-coast family (**pace4k** belongs to the ocean family). Same conventions as Table 2; the annual maps are targets, the rest inputs.

so one codec serves optical and radar, with a same-place-different-date contrastive term so the token carries state, not texture. It emits $k = 4\text{--}8$ tokens of 64 dimensions, cached once, reused by every anchor whose cone holds the tile; it sees one date, and the cone codec supplies time. Tile size is a knob: 640 m tiles at seventy clear dates a year are ~ 20 TB of tokens a year globally, 2.56 km tiles sixteen times less. Imagery is streamed once to train and once to encode, never kept.

3.3 Biosphere observations

The living surface is observed by few instruments and fitted by many maps. The measurements are the laser footprints (GEDI [35] to $\pm 51.6^\circ$, ICESat-2 [36] beyond), which alone see a canopy in three dimensions; fluorescence soundings [37], emitted by photosynthesis itself; the towers' eddy-covariance carbon flux [38], the long in-situ record; leaf area every eight days; fire detections; and 3.7 billion dated, located species occurrences [39] — sparse and biased, and the only global record of where life is. The annual maps — forest loss [40], canopy height [41, 42, 43], biomass [44], the leaf calendar — are each fitted to one of those measurements and are admitted as targets: what a user asks for, at the map's own footprint and year. Nothing on land advects; the drivers of every biosphere channel are the atmosphere and the soil, which every token already carries.

3.4 Training, validation and test periods

Development years 2009, 2017, 2023 are held out of training, climatology and every statistic; the terminal test scores a model trained through 2020 once on 2021–2024. An example is admissible only if every bin its forward pass can see — every past token, every future target — is a training bin; a brute-force count over a large anchor draw certifies this before a run, and a non-zero count refuses it. Longitude holes are held out too, so extrapolation is scored apart from interpolation.

4 Training and scaling

Data is priced in observed values and their independence, not bytes. A 0.25° cell is correlated over hundreds of kilometres and, at the surface, months, so a basin's 10^9 values are $10^6\text{--}10^7$ independent ones — the regime in which a small model equals a large one. The globe adds regimes, the spheres

rung	values	tokens	ceiling	what it takes
one basin, 0.25° pentads, 42 channels	2.5 B	0.04 B	0.12 B	the pilot’s tensor
global pentads, four groups	13.8 B	0.21 B	0.7 B	built (§3.1)
+ the shared channels of every sphere	91 B	1.4 B	4.5 B	ERA5, radiometers, scatterometers, gravimetry, soil moisture
daily cadence, all spheres	455 B	7.0 B	23 B	a 1.3 TB tensor and a streaming loader
kilometre tier	3.5×10^5 B	5.4×10^3 B	—	the token cache; compute-bound

Table 4: The data ladder in observed values; tokens at 65 channels each; ceiling at one parameter per twenty values [13], an upper bound because values are correlated.

add physics, daily cadence is mostly the same sample five times, the kilometre tier adds new variance. Order: globe and spheres, then footprint, cadence last (Table 4).

Each rung is bought on the previous rung’s read-out under the frozen protocol with the null ladder beside it; a rung whose held-out minimum arrives inside 2,000 steps is over-parameterised and the next is not bought. L1: a 7–125 M codec on the 0.25° pentad grid, fixed against learned geometry. L2: 0.3–1 B codec and forecaster with the tier-P stores; the width ladder is re-measured here, the one test that separates data-limited from compute-limited. L3: 1–4.5 B, sphere-routed experts, all spheres: do land and atmosphere improve the ocean’s rolled skill and vice versa. L4: daily cadence, the first compute-bound rung. L5: the kilometre tier, after L4 is positive. Compute is $6NT$ per epoch; a 1 B codec over the 0.25° pentad grid is ~ 30 RTX-4090-hours at four epochs, and the aperture and ring cost one multiply per token. The gather is host-bound and batched by latitude row. Every artefact is written incrementally to durable storage while a run lives. A configuration is a named recipe; a resumed run derives its architecture from its checkpoint. One seed is a direction, never a level; the first result at a new tier buys its replication, and a probe-scored claim needs three seeds.

5 Evaluation methodology

5.1 Protocol and baselines

Development holdout as in §3.4; terminal test once. Per lead: mean-squared skill against climatology, raw and damped persistence; anomaly correlation; amplitude ratio; per channel and scope (trained and held-out longitudes); three starts per held-out year; block-bootstrap intervals over (year, start) [17]. The null ladder beside every read-out: climatology, persistence, damped persistence (per-cell AR(1) on training years), the LIM in data and in embedding space [15, 16], an analogue forecast, the previous rung. On the atmospheric channels at leads up to fifteen days the external baseline is FGN [21], a data-driven ensemble whose released checkpoints (WeatherNext 2) are run here from ERA5 initial conditions over 2023, its own test year and a held-out year here; its 6-hourly 0.25° members are averaged to the pentad bins and the 1° cells the atmospheric group holds, so both models are scored on the same support. Scores: fair CRPS [18] of the ensemble, ensemble-mean RMSE and spread–skill ratio, per channel and lead, and CRPS pooled over 10° blocks. Configurations are compared on the dense field; the read-outs are checks. A winner is replicated with a second seed before it is written up; a negative result is reported like a positive one.

5.2 Pre-registered comparisons

(1) **Hourglass against snapshot codec**, both frozen, same forecaster: one-step ratio, future-cone families against their bars, the T1–T2 gap.

(2) **Geometry**, same tensor, codec size and steps, three seeds each: **A0** fixed (control and evaluation geometry); **A1** learned per channel, hierarchical, from A0’s initialisation; **A1g** learned per group; **A2** as A1 with the ocean-surface drift primed at $-\mathbf{u}_{\text{clim}}\Delta t$. Read-outs, fixed now: **R1** T1 loss per lead against its bar; **R2** the T1–T2 gap; **R3** per-channel drift against its expected carrier over the world ocean where the carrier exceeds 0.2 m s^{-1} — currents against $-\mathbf{u}_{\text{clim}}$, height’s zonal drift positive (source to the east), ice against the reversed wind turned 5–30°, SST against the reversed wind at lags 0–1, deep levels against the reversed reanalysis velocity at their depth; **R4** at an anchor north of the 26.5°N

array, surface drift south-west up the Florida Current and the drift of the $\geq 1,000$ dbar channels north up the deep western boundary current, with attention on deep profiles north and shallow south; **R5** the target-variance flag. Verdict: A1 is adopted if it beats A0 on R1 at every lead ≤ 3 , paired at all seeds, with R5 clean; A1 tied with A1g keeps the per-group form as the cheaper equal; A2 beating A1 with A1 at A0 means the gradient cannot find upstream and the geometry is set from climatology; none winning keeps A0 and points to a per-example cone. First ablation after: A1 without the ring.

(3) **Forecaster**: backbone plus residual against the transformer alone; direct multi-lead against rolled one-step; log-spaced against dense lags.

(4) **Footprint and tiers**: coarse groups as offset tokens with true footprints against nearest-cell copies; the stores of Table 2 and the point families with against without, per read-out — El Niño by Niño-3.4 at six months against warm-water-volume persistence; carbon by held-out surface CO₂ against a climatological null; overturning by the velocity loss with drifter tokens against without; SST by the level-3 cell loss with drifter and ship tokens against without; the land biosphere by held-out tower carbon flux against a per-site day-of-year climatology and damped persistence, and by canopy height at held-out laser footprints against the 2020 map and the footprint’s previous value, with fluorescence at held-out soundings as the dense check — and per store on the read-out it was admitted for; the coastal tile codec with against without tile tokens, three seeds; AlphaEarth’s cache in place of the tile codec, scored on the canopy read-out as well, since that embedding was fitted to the same laser.

(5) **Scale**: the width ladder at L2; sphere-routed experts against a dense model of equal active parameters at L3.

(6) **Weather**: the shared atmospheric channels — 2m temperature, 10m wind, surface pressure, precipitation — at leads of one, two and three pentads against FGN on 2023, with damped persistence and climatology beside both. The forecaster is not a weather model: it reads pentad means and predicts pentad means, and FGN is scored on the same means. The comparison says how much of the medium-range atmosphere a model built for the coupled system at pentad cadence gives up, and whether the ocean, land and biosphere stores buy any of it back (arms with and without them).

Build order: geometry, task mix, aperture, maps and harden step, forecaster, tiers; no comparison changes it.

6 Results

Reserved. Nothing here has been trained; the tables fix the columns now. Each cell will carry its seed count and spread.

6.1 Encoder: hourglass against snapshot

codec	one-step ratio	lead 1 / bar	lead 3 / bar	lead 6 / bar	T1–T2
snapshot (waist only)	—	—	—	—	—
hourglass, fixed (A0)	—	—	—	—	—

Table 5: Comparison (1). Ratios: held-out loss over the predict-the-mean bar; one-step ratio: the forecaster’s z -space error over persistence.

6.2 Cone geometry

arm	R1 lead 1	R1 lead 2	R1 lead 3	R2	R3	R5
A0 fixed	—	—	—	—	0 by construction	—
A1 per channel	—	—	—	—	—	—
A1g per group	—	—	—	—	—	—
A2 primed	—	—	—	—	—	—
A1 without ring	—	—	—	—	—	—

Table 6: Comparison (2), three seeds per arm, paired. R4 is a figure.

6.3 Forecaster and the null ladder

lead (pentads)	1	3	8	21	34	55	73
persistence	—	—	—	—	—	—	—
damped persistence	—	—	—	—	—	—	—
LIM, data space	—	—	—	—	—	—	—
LIM, embedding space	—	—	—	—	—	—	—
analogue	—	—	—	—	—	—	—
forecaster, backbone + residual	—	—	—	—	—	—	—
without backbone	—	—	—	—	—	—	—
one-step, rolled	—	—	—	—	—	—	—
dense lags	—	—	—	—	—	—	—

Table 7: Comparison (3) and the null ladder: $\text{MSSS}_{\text{clim}}$ on the dense field, world ocean, per lead; the same table for $\text{MSSS}_{\text{damped}}$, ACC and amplitude, per channel and scope.

6.4 Observation stores per read-out

read-out	score	null	without stores	with stores
El Niño	Niño-3.4 correlation at six months	warm-water-volume persistence	—	—
carbon sink	held-out surface- CO_2 error	climatological null	—	—
overturning	velocity loss / bar; transport r at 26.5°N	drifter-free cone; damped persistence	—	—
SST	level-3 cell loss / bar	drifter- and ship-free cone	—	—
land biosphere	tower flux error / bar; canopy height error at laser footprints	site climatology, damped persistence; the 2020 map, previous value	—	—

Table 8: Comparison (4): the stores per read-out, and the footprint fields, against the same cone without them. The biosphere row is also scored with AlphaEarth’s cache in place of the tile codec.

6.5 Scaling

width	held-out minimum step	one-step ratio	mean ACC, leads 1–73	mean $\text{MSSS}_{\text{damped}}$
—	—	—	—	—

Table 9: Comparison (5): the width ladder at L2, one row per width.

6.6 Weather against FGN

channel	lead	fair CRPS			ensemble-mean RMSE		
		FGN	ours	damped	FGN	ours	damped
2 m temperature	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—
10 m wind speed	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—
surface pressure	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—
precipitation	1	—	—	—	—	—	—
	2	—	—	—	—	—	—
	3	—	—	—	—	—	—

Table 10: Comparison (6): the atmospheric channels on 2023, pentad means on the 1° cells, leads in pentads (5, 10, 15 days). FGN: the released WeatherNext 2 checkpoints from ERA5 initial conditions, 64 members; ours: the seed ensemble of §2.6; damped: per-cell AR(1). The same table for spread-skill ratio and 10°-pooled CRPS, and the arms with and without the fine stores.

7 Related work

Global data-driven weather models [19, 20, 22, 21] and their hybrids with a dynamical core [23], atmospheric foundation models [24, 25], end-to-end systems reading observations [26], ocean emulators [27, 28] and coupled emulators [29] read one grid, one variable set and one reach. This design reads every source at its own footprint through one token, samples a per-channel cone, and predicts through a query carrying the footprint of the answer. Speed times time as a receptive field has precedents: an ocean emulator drops lateral boundaries because eddies travel a bounded distance over the horizon [28]; a neural semi-Lagrangian layer gathers at $\mathbf{x} - \Delta t \mathbf{u}$ with one velocity shared across transported modes [30]; deformable attention learns offsets per query and head with a learning-rate multiplier [8]. Truncation by memory, a cone per channel, and a learnable ellipse with a hierarchical per-channel drift are, to the authors’ knowledge, not stated as a principle elsewhere; per-variable tokenisation [24, 25] changes what a variable is embedded as, not where it is read from. The architecture is a masked autoencoder [9] with Perceiver cross-attention [10]; AlphaEarth Foundations [2] is the closest embedding-field model, over land at annual cadence, and enters here as a token cache; geospatial tile models [31] embed imagery. Global canopy-height maps [41, 42, 43] regress one year of laser footprints from imagery, one map per year; here the footprints are targets in their own coordinates and the maps are one more target, so the forecast of a canopy is scored against the laser, not against a map. Frozen backbones with shallow read-outs [32] are the evaluation stance, judged by what a forecaster predicts from the representation rather than by a current-state probe. The LIM [15, 16] and the interannual-to-decadal verification framework [17] are the nulls and scores; the decoupled objective follows [11], fair CRPS [18], the four-epoch rule [12], sparse experts [14]. The physics of the per-channel carriers is from [3, 4, 5, 6]; the overturning array [33] and Argo [34] enter as observations in their own coordinates.

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A Data and code availability

Code: <https://github.com/blauewelt/earth> — geometry (`ml/cone.py`), sampler (`ml/cone_sampler.py`), codec and masking plan (`ml/cone_codec.py`, `ml/train_cone.py`), point-store reader (`ml/family10_store.py`), builders (`ml/build_family7.py`, `ml/build_family8_argo.py`, `ml/build_family10_stores.py`). Data: the 0.25° pentad grid (`family7_global025_pentad_l2`), the Argo store, the point-and-track stores and the fine-granularity families as built, in the Hugging Face dataset `chfrank/earth-tensors`, each with a registry of checksums, sources and licences; private-track stores are listed there and held privately. Design notes: `ml/paper/notes/`; plans and verdict rules: `ml/plans/`. Sources: GLORYS12 and GREP (Copernicus Marine), OISST v2.1 (NOAA NCEI), NCEP/NCAR Reanalysis-1 (NOAA PSL), the Roemmich–Gilson Argo climatology (Scripps; Argo data from the International Argo Program), OC-CCI v6.0 (ESA CCI, PML), RAPID-MOCHA-WBTS (NERC, NSF, NOAA), the Global Drifter Program and GTMBA (NOAA AOML, PMEL), SOCAT, Copernicus Marine along-track sea level, GEDI (NASA LP DAAC, ORNL DAAC), ICESat-2 ATL08 (NSIDC), TROPOMI fluorescence (Caltech, ESA), FLUXNET, GBIF, MODIS and VIIRS land products (LP DAAC), GLAD, ETH and Meta canopy height, ESA CCI Biomass, Hansen GFC, PACE (NASA OB.DAAC), and the sources named per store in the registries.